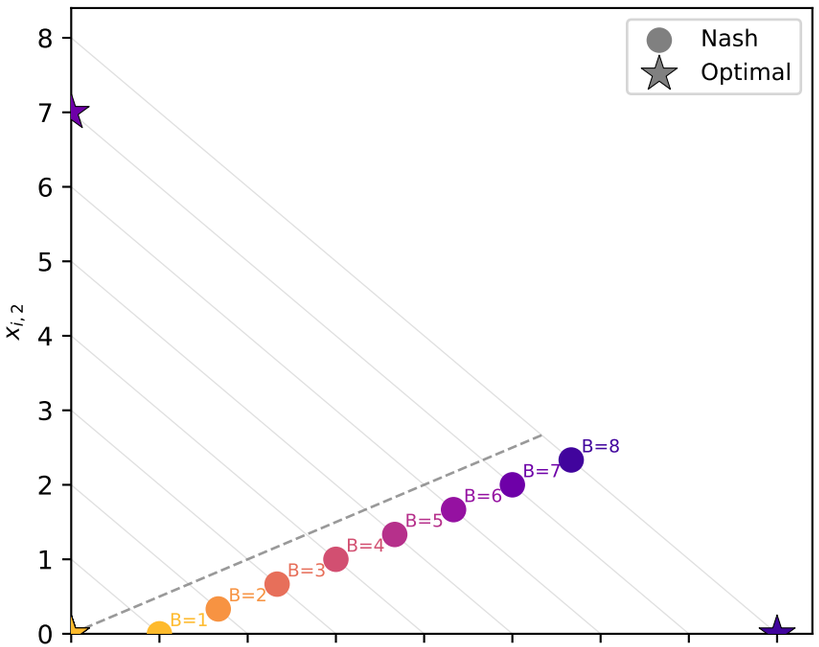
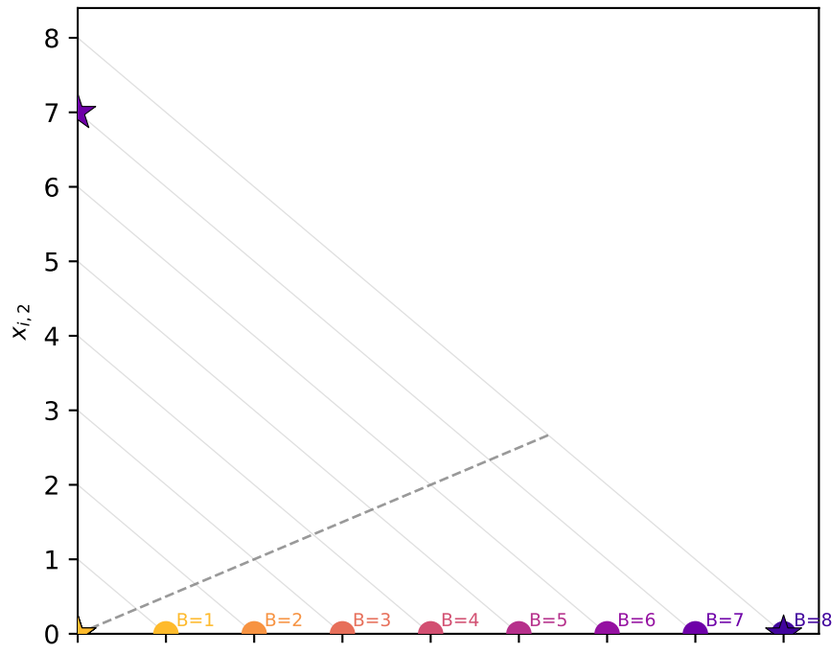
Nash (circles) vs Optimal (stars): $K = 2$, $w_1 = 2/3$, $w_2 = 1/3$, $N = 8$ firms
(dashed = proportional ray)

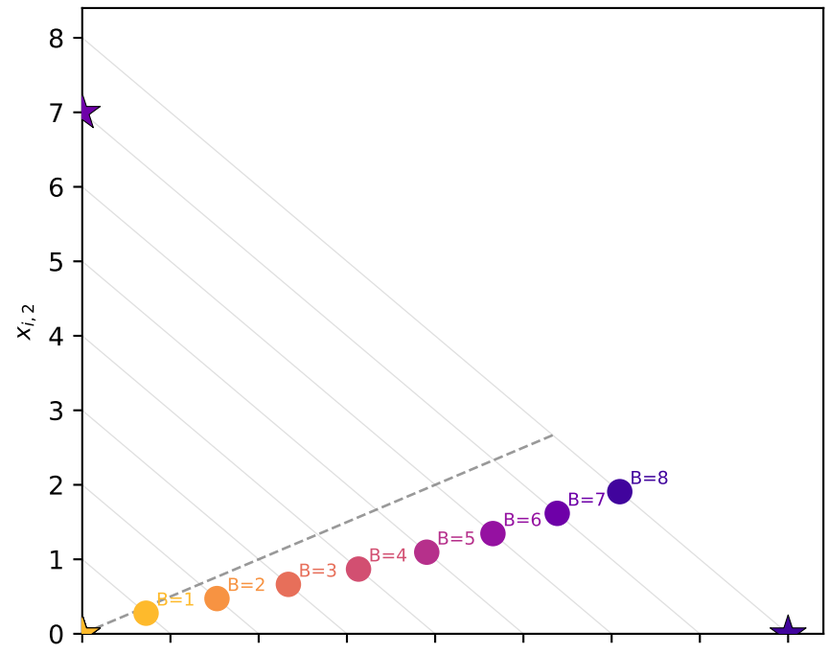
$f(x) = log(x + 1)$ **Nash=1.666 | Opt=2.158 (PoA=0.77)**



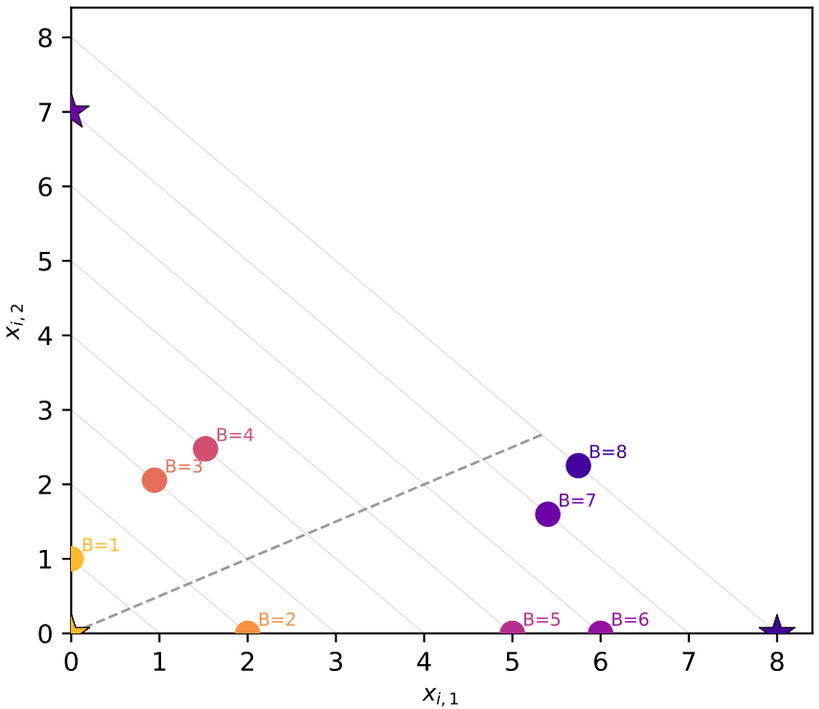
$f(x) = log(x + 10)$ **Nash=2.694 | Opt=2.871 (PoA=0.94)**



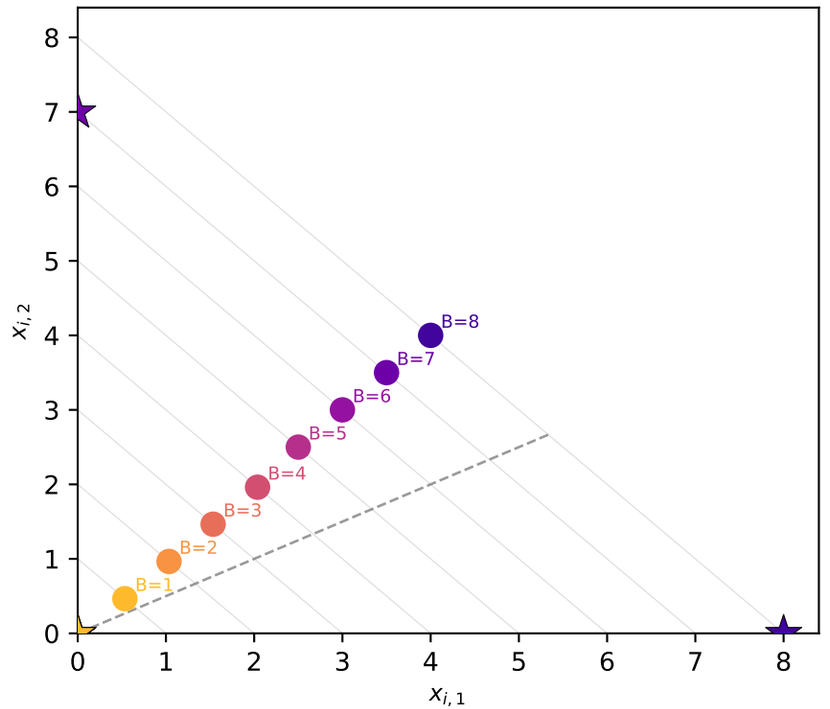
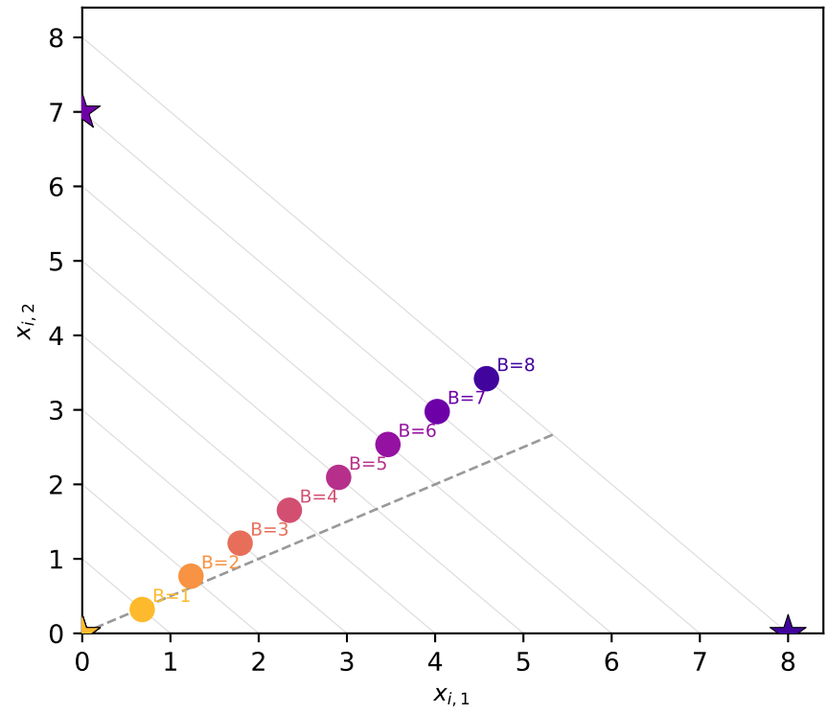
$f(x) = sqrt(x)$ **Nash=2.106 | Opt=2.768 (PoA=0.76)**



$f(x) = x^{0}.99$ **Nash=4.747 | Opt=7.512 (PoA=0.63)**



$f(x) = 1 - e^{-10x}$ **Nash=1.000 | Opt=1.000 (PoA=1.00)**



$f(x) = 1 - (1 + x)^{-2}$ **Nash=0.962 | Opt=0.987 (PoA=0.97)**